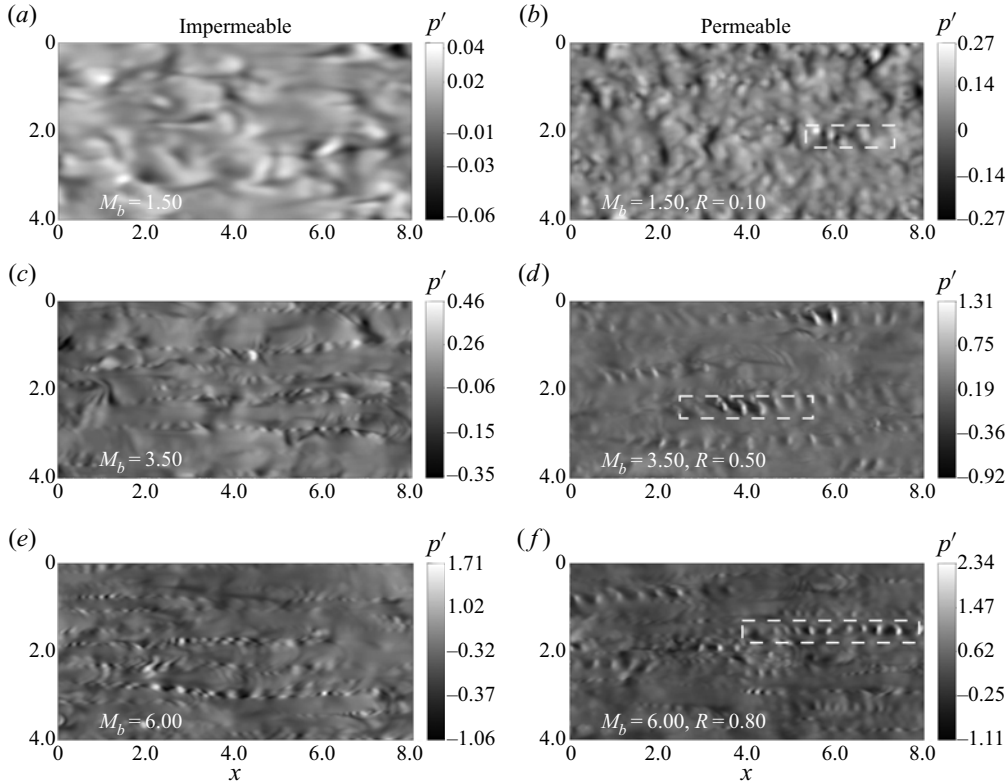

Figure 16. Top–down view of the pressure fluctuations on the bottom wall. Left column shows the LES impermeable-wall results, right column shows the permeable-wall cases presented in figure 15, with the white box showing the region where filtered pressure fields are extracted.

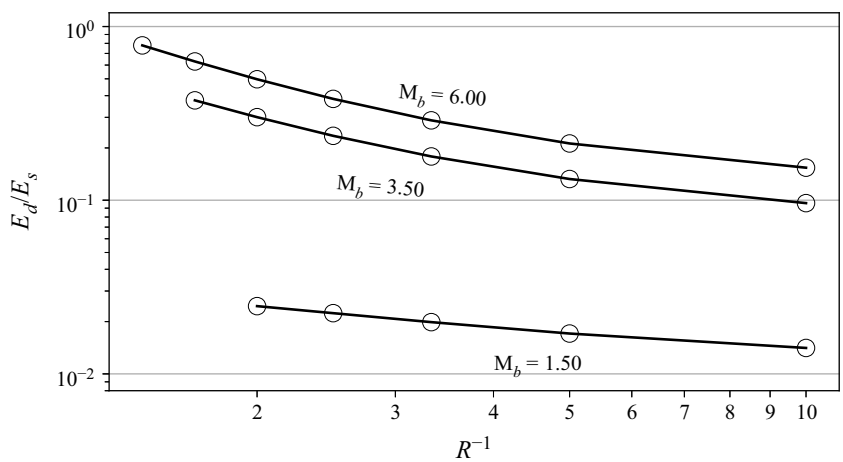

Figure 17. Ratio of the integrated norm of dilatational part $\hat{\boldsymbol{v}}_d$ and solenoidal part $\hat{\boldsymbol{v}}_s$ of the perturbed field, E_d/E_s, defined in (7.3) vs the inverse of the wall resistance. Results are obtained by applying the Helmholtz–Hodge decomposition to the velocity eigenfunctions of the unstable mode. Only resistance values below R_{cr} are shown here, with $\lambda_x = 0.47$.

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components, the ratio of the integrated norm

$$\frac{E_d}{E_s} = \frac{\int_0^2 |\mathrm{Re}\{\hat{\boldsymbol{v}}_d \mathrm{e}^{-\alpha c}\}|^2 \,\mathrm{d}y}{\int_0^2 |\mathrm{Re}\{\hat{\boldsymbol{v}}_s \mathrm{e}^{-\alpha c}\}|^2 \,\mathrm{d}y} \tag{7.3}$$

is evaluated for the normal velocity component for cases with $R < R_{cr}$ for all Mach numbers. Results are shown in Figure 17 for wavenumbers (wavelengths) associated with those appearing in LES. At each Mach number, the dilatational field occupies a discernible amount of energy once the instability is triggered, and is gradually weakened as permeability increases. The perturbed flow fields exhibit a more hydrodynamic nature as the response of the impedance wall to the overlying turbulent field becomes stronger. In addition, the same (non-dimensionalized) permeability tends to yield a perturbed field with higher portion of dilatational energy at higher Mach number.

8. Conclusion

To examine the instability phenomenon of high-speed turbulent flows over porous media, LES of compressible turbulent channel flows over walls of various impedance have been performed at three bulk Mach numbers:$M_b = 1.50$, 3.50 and 6.00. The bulk Reynolds number Re_b is tuned to achieve a similar viscous Reynolds number $Re_\tau^* \approx 220$ to ensure a resemblance of near-wall turbulence structures over impermeable walls. The TDIBC based on the ADE method is applied to the bottom wall of the channel. The three-parameter complex impedance model with a resonating frequency tuned to the large-eddy turn-over frequency of the flow is adopted.

Results show that a finite wall permeability has a significant effect on the near-wall turbulence. Increasing mean wall-shear stress has been observed when the permeability increases, as a statistical result of the more frequently appearing extreme instantaneous high wall-shear stresses compared with the impermeable-wall baseline case. At sufficiently low resistance, the visualization using the Q-criterion captured streamwise-travelling waves entailing high–low alternating wall pressure patterns. Such wave structures contribute to the production of TKE in the near-wall region at all Mach numbers, with the contribution weakening as the Mach number increases.

LSA using the turbulent background base flow confirms that the aforementioned instability is triggered by the finite wall permeability and is trapped near the bottom wall, when the resistance is below a threshold defined as critical resistance R_{cr}. The pressure mode shape extracted from LSA agrees well with those from the filtered LES simulations. The critical resistance R_{cr} is shown to be sub-linearly proportional to the bulk Mach number M_b, indicating that less permeability is required to trigger the instability in high Mach number flows. By applying Helmholtz decomposition of the perturbed velocity field obtained from LSA, it is shown that the occupation of the dilatational field gradually disappears as the response of the impedance wall to the overlying flow gets stronger, i.e. as R decreases.

For future work, it would be desirable to have experimental studies to confirm the behaviour and effect of the wave structures in high-speed flows. In addition, in the current study, the resonating frequency ω_{res} was picked to couple with the overlying turbulence, which is not necessary the case in real applications, neither are the resistance values

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selected in this paper. Thus, future studies might also include parametric studies of ω_{res} and R, both of which can in fact have spatial variations.

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Appendix A. Semi-implicit time advancement scheme

In this section, some details of the semi-implicit time advancement scheme used in this work are presented. While this scheme is not novel *per se*, the authors have found that its adoption was necessary to stabilize calculations with strong flow responses such as the ones obtained for low resistance values $R < 0.5$ entailing strong wall-normal velocities near the boundaries. For simplicity, a shorthand notation of the governing equations is used here to illustrate the method

$$\boldsymbol{U}_t = \boldsymbol{A}(\boldsymbol{U}) + \boldsymbol{B}(\boldsymbol{U}), \tag{A1}$$

where $\boldsymbol{U}$ is the solution vector; $\boldsymbol{A}(\boldsymbol{U})$ and $\boldsymbol{B}(\boldsymbol{U})$ are explicit and implicit flux vectors, respectively. Then from time step n to $n+1$, the kth stage of an explicit Runge–Kutta scheme can be written in the following form:

$$\left.\begin{aligned} \frac{\boldsymbol{U}^k - \boldsymbol{U}^{k-1}}{\Delta t} &= \beta_k[\boldsymbol{B}^k + \boldsymbol{B}^{k-1}] + \gamma_k \boldsymbol{A}^{k-1} + \zeta_k \boldsymbol{A}^{k-2}, \quad k = 1, 2, \ldots, s \\ \boldsymbol{U}^0 &= \boldsymbol{U}^n, \quad \boldsymbol{U}^s = \boldsymbol{U}^{n+1} \end{aligned}\right\}, \tag{A2}$$

where s represents the total number of stages of the explicit scheme, which is 3 in current method; α_k and β_k are coefficients given below

$$\left.\begin{array}{ccc} \beta_1 = 4/15 & \beta_2 = 1/15 & \beta_3 = 1/6 \\ \gamma_1 = 8/15 & \gamma_2 = 5/12 & \gamma_3 = 3/4 \\ \zeta_1 = 0 & \zeta_2 = -17/60 & \zeta_3 = -5/12. \end{array}\right\} \tag{A3}$$

Then the problem is to approximate the flux $\boldsymbol{B}^k$, which is unknown at stage k. A linearization in time leads to

$$\left.\begin{aligned} \boldsymbol{B}^k &= \boldsymbol{B}^{k-1} + \left(\frac{\partial \boldsymbol{B}}{\partial \boldsymbol{U}}\right)^{k-1} \Delta \boldsymbol{U}^{k-1} + O(\Delta t^2), \\ \Delta \boldsymbol{U}^{k-1} &= \boldsymbol{U}^k - \boldsymbol{U}^{k-1} \end{aligned}\right\}. \tag{A4}$$

To iteratively solve the system, sub-iterations can be incorporated into each stage to ensure

$$\lim_{l \to \infty} \boldsymbol{U}^l = \boldsymbol{U}^k. \tag{A5}$$

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The delta formulation is required here by replacing the terms at time level k with those at level l (not the coefficients)

$$\frac{\boldsymbol{U}^{l}-\boldsymbol{U}^{k-1}}{\Delta t}=\beta_k\{\boldsymbol{B}^{l}+\boldsymbol{B}^{k-1}\}+\gamma_k\boldsymbol{A}^{k-1}+\zeta_k\boldsymbol{A}^{k-2}. \tag{A6}$$

Linearizing $\boldsymbol{B}^{l}$ and rearranging (A6), one can arrive at

$$\left\{I-\Delta t\beta_k\left(\frac{\partial\boldsymbol{B}}{\partial\boldsymbol{U}}\right)^{l-1}\right\}\Delta\boldsymbol{U}^{l-1}=\boldsymbol{U}^{k-1}-\boldsymbol{U}^{l-1}+\Delta t\beta_k\boldsymbol{B}^{l-1}$$
$$+\Delta t(\beta_k\boldsymbol{B}^{k-1}+\gamma_k\boldsymbol{A}^{k-1}+\zeta_k\boldsymbol{A}^{k-2}). \tag{A7}$$

The current implementation treats all wall-normal convective fluxes implicitly. For the viscous flux, only those containing second-order wall-normal derivatives are included. The detailed formulation, including the Jacobians, can be found in Nagarajan (2004). A test case will be shown in B.1 for validation.

Appendix B. Verification and validation of TDIBC implementation

B.1. *Impedance tube test problem*

A one-dimensional impedance tube problem is used to validate the implementation of TDIBC in CFDSU. A broadband wave pulse with initial shape is defined as follows on $y\in[0,2]$ (at $t=0$):

$$p'(y,t)=\tfrac{1}{2}A\exp(-\alpha k^2(y-1-t)^2)\cos[2\pi k(y-1-t))], \tag{B1}$$

$$v'(y,t)=\pm p'(y,t), \tag{B2}$$

where '+' and '−' signs are used for initially up-travelling and down-travelling waves, respectively. With the assumption of linear advection of the waves, the solution is given by

$$v^{\pm}(y,t)=\frac{A}{4k}\sqrt{\frac{1}{\alpha\pi}}\int_{-\infty}^{+\infty}\frac{1-\hat{Z}(\omega)}{1+\hat{Z}(\omega)}\exp(\pm\mathrm{i}\omega(y-t-1))$$
$$\times\left[\exp\left(-\frac{(\omega/k+2\pi)^2}{4\alpha}\right)+\exp\left(-\frac{(\omega/k-2\pi)^2}{4\alpha}\right)\right]\mathrm{d}\omega. \tag{B3}$$

The above integral is performed numerically for $t=2.0$ at which the wave hits the wall once. The free parameters are selected to be $\alpha=0.8$, $k=7.0$, $A=1e-7$. This problem was tested by using ADE method with various spatial and temporal resolutions, with the spatial and temporal convergence study given in figure 18. Figure 19 presents the solution of an initially down-travelling wave, which agrees well with the numerically obtained analytical solution.

B.2. *Validation of TDIBC applied to a turbulent channel flow*

To check if the TDIBC has been correctly implemented in the simulation, the pressure–velocity cross-spectrum is examined, derived as

$$\hat{p}/\rho_w a_w=-\hat{Z}(\omega)\hat{v}\implies\hat{p}\hat{p}^{\star}/\rho_w a_w=-\hat{Z}(\omega)\hat{v}\hat{p}^{\star}. \tag{B4}$$

The phase and magnitude of $\hat{p}\hat{p}^{\star}/\rho_w a_w$ and $-\hat{Z}(\omega)\hat{v}\hat{p}^{\star}$ are compared in figure 20 for cases at $M_b=1.50$. Good agreements are achieved in the magnitude of the spectrum.

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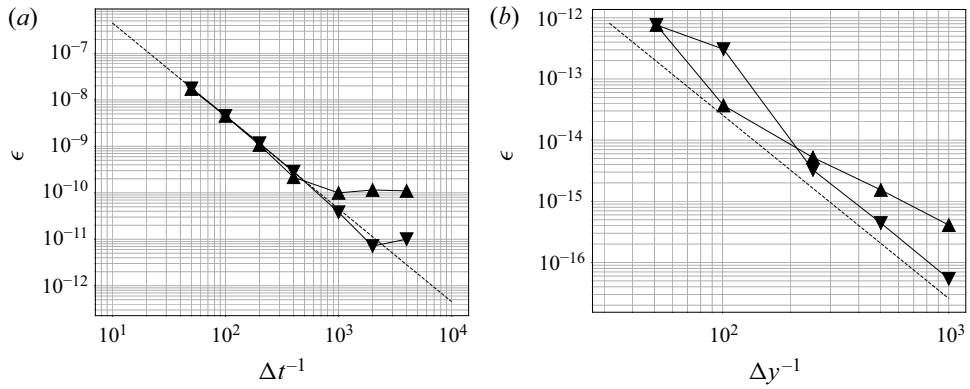

Figure 18. The spatial and temporal convergence study for the impedance tube test problem with initially up-travelling (▲) and down-travelling (▼) waves. The dashed line represents a second-order convergence in (*a*) and a third-order convergence in (*b*). For the spatial convergence study, all cases are run with $CFL \approx 0.005$. For the temporal convergence study, 1001 grid points are used in the y direction.

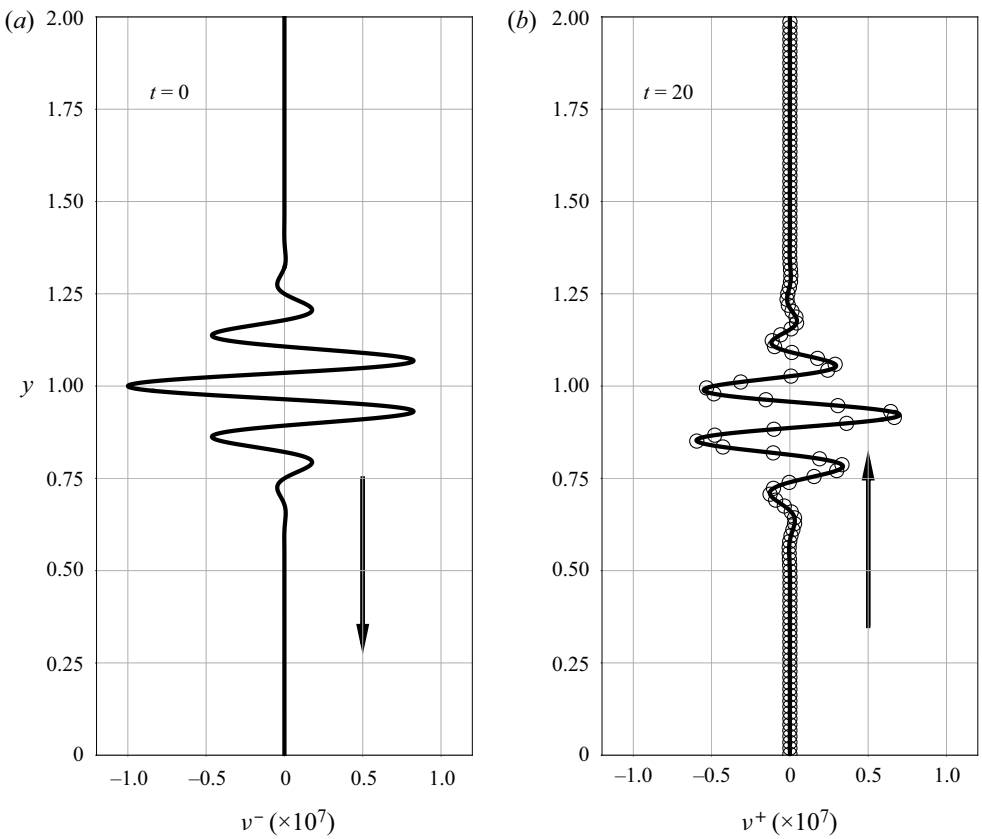

Figure 19. The impedance tube test case with an initially down-travelling broadband wave pulse. TDIBC is applied at the bottom wall. Initial condition at $t = 0$ (*a*) and result at $t = 2$ (*b*). Analytical solution at $t = 2$ (unfilled circles) is also shown (plotted every four points).

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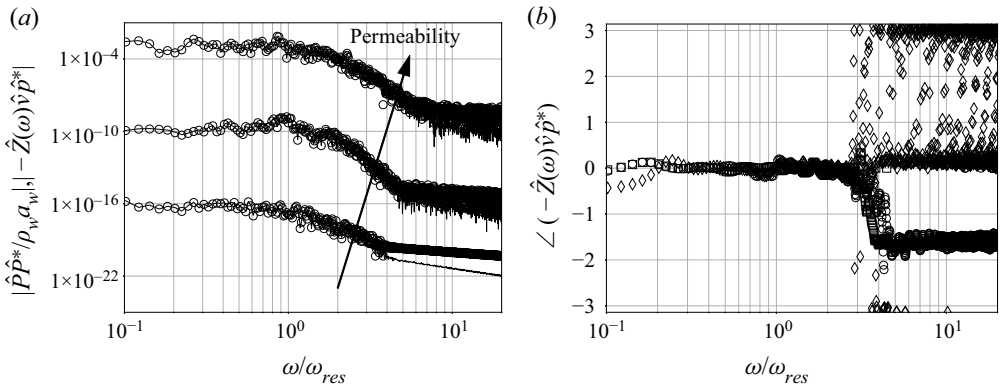

Figure 20. Magnitude (*a*) of the pressure–velocity cross-spectrum $\hat{v}\hat{p}^\star$ for permeable-wall cases at $M_b = 1.50$. Magnitude of the data is shifted for clarity with the arrow showing the direction of increasing permeability (decreasing R). Quantities presented are $|\hat{p}\hat{p}^\star/\rho_w a_w|$ (unfilled circles) and $|-\hat{Z}(\omega)\hat{v}\hat{p}^\star|$ (black solid line), which should match if the TDIBC is correctly implemented. The phase (*b*) of $|-Z(\omega)\hat{v}\hat{p}^\star|$ is also shown for $R = 0.10$ (unfilled circles), $R = 0.50$ (unfilled diamonds) and $R = 1.00$ (unfilled squares). Results match the definition in (1.1).

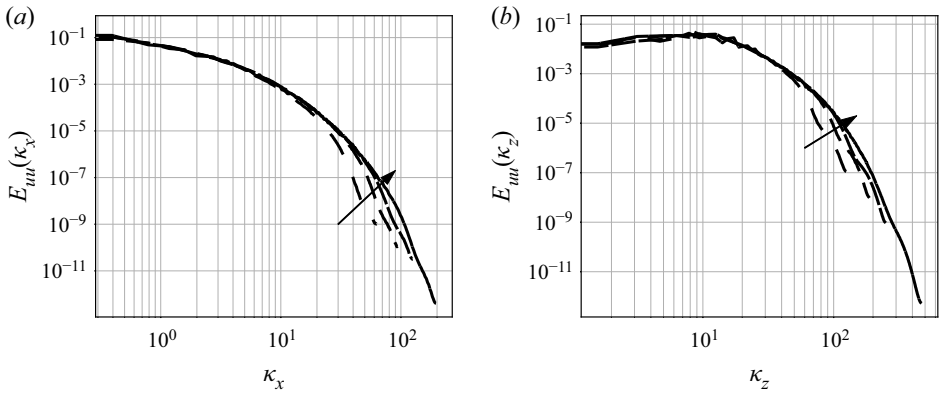

Figure 21. One-dimensional spectrum of streamwise velocity at $y^* \approx 15$ in $M_b = 6.00$ DNS case, with arrow showing increasing resolution ($N_x \times N_y \times N_z$): $320 \times 200 \times 240$, $480 \times 250 \times 360$, $640 \times 300 \times 480$ and $1000 \times 300 \times 600$.

Slight deviations in phase have been observed near the resonating frequency, which decrease as the wall permeability drops.

Appendix C. Turbulent channel flow over impermeable walls

The grid-refinement study is performed for the DNS case at $M_b = 6.00$ and the spatial spectra are given in figure 21. The two-point correlations are presented in figure 22. Two locations are selected to present the data: at $y^* \approx 15$ and the channel centre. The streamwise and spanwise correlations of velocity components all approach zero when the separation is close to half of the domain size in each direction, implying the satisfaction of the simulation box size. For the pressure correlation, all the data have a good decaying trend except in the spanwise direction at the channel centre plane, which gives a value

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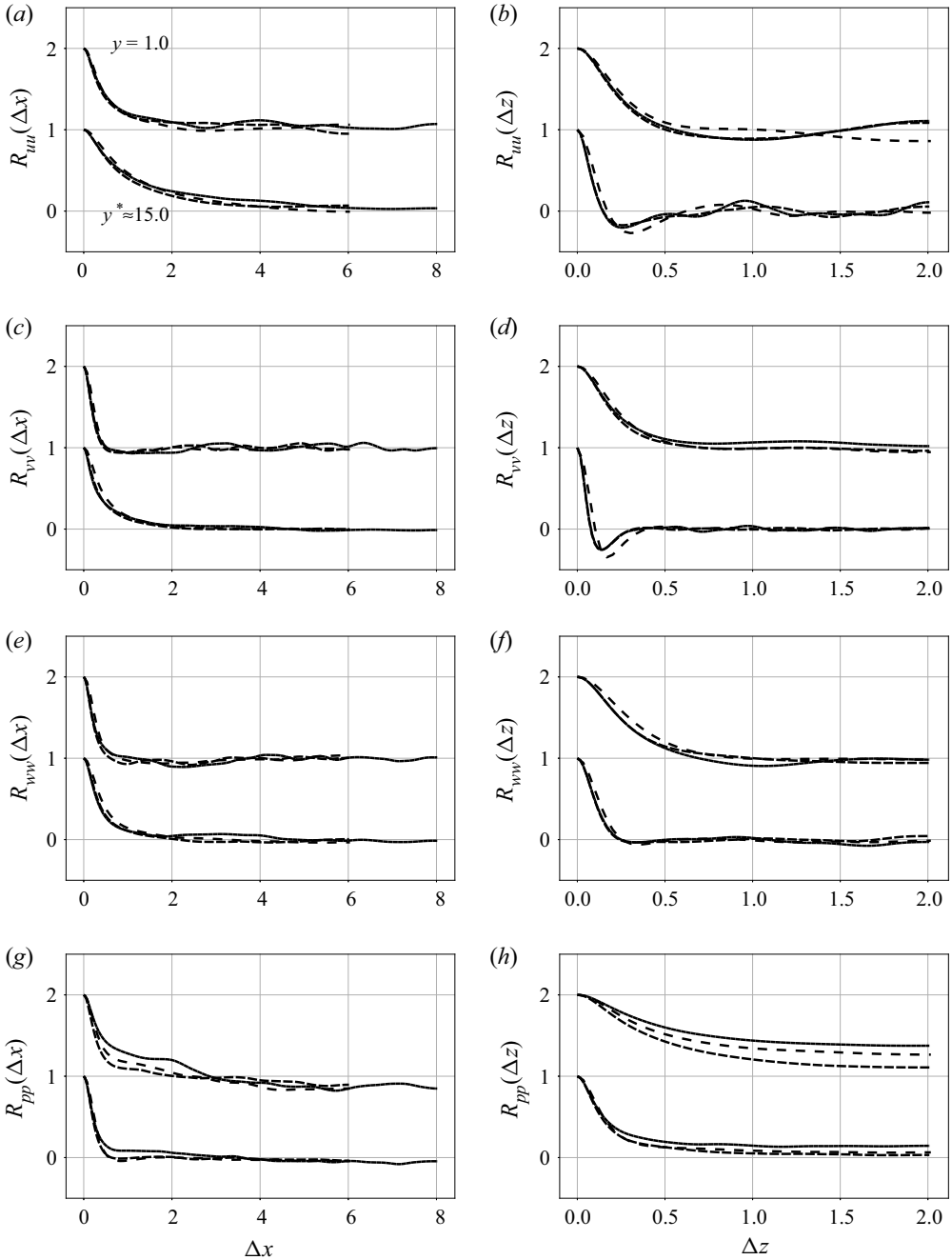

Figure 22. Two-point correlations in homogeneous directions of impermeable-wall cases at different Mach numbers: $M_b = 1.50$ (black dotted line), $M_b = 3.50$ (black dotted line), $M_b = 6.00$ (black solid line). Correlations are taken at channel centre $y = 1.0$ (shifted up) and buffer layer $y^* \approx 15$ for each Mach number.

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around 0.1 near $\Delta z = 2$. By inspecting of the flow data, it is found that the fluctuation of pressure near the centre plane is very small, which causes the correlation curve to plateau.

Appendix D. Linear stability

D.1. *Solver validation*

The two-dimensional linearized Navier–Stokes equations and the equation of state, written in terms of the primitive variables are

$$\frac{\partial \rho'}{\partial t} + \frac{\partial \rho' \bar{u}_j}{\partial x_j} + \frac{\partial \bar{\rho} u_j'}{\partial x_j} = 0 \tag{D1}$$

$$\bar{\rho}\left(\frac{\partial u_i'}{\partial t} + \bar{u}_j \frac{\partial u_i'}{\partial x_j} + u_j' \frac{\partial \bar{u}_i}{\partial x_j}\right) + \rho' \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{\partial p'}{\partial x_i} + \frac{1}{Re_a}\frac{\partial}{\partial x_j}\left[\bar{\mu}\left(\frac{\partial u_i'}{\partial x_j} + \frac{\partial u_j'}{\partial x_i} - \frac{2}{3}\delta_{ij}\frac{\partial u_k'}{\partial x_k}\right) + \mu'\left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} - \frac{2}{3}\delta_{ij}\frac{\partial \bar{u}_k}{\partial x_k})\right)\right] \tag{D2}$$

$$\begin{aligned}\bar{\rho}\left(\frac{\partial T'}{\partial t} + \bar{u}_j \frac{\partial T'}{\partial x_j} + u_j' \frac{\partial \bar{T}}{\partial x_j}\right) + \rho' \bar{u}_j \frac{\partial \bar{T}}{\partial x_j} &= (\gamma - 1)\left(\frac{\partial p'}{\partial t} + \bar{u}_j \frac{\partial p'}{\partial x_j} + u_j' \frac{\partial \bar{p}}{\partial x_j}\right) + \frac{1}{PrRe_a}\frac{\partial}{\partial x_j}\left(\bar{k}\frac{\partial T'}{\partial x_j} + k'\frac{\partial \bar{T}}{\partial x_j}\right) \\ &\quad + \frac{\gamma - 1}{Re_a}\left\{2\bar{\mu}\left[\frac{\partial \bar{u}_k}{\partial x_j}\frac{\partial u_k'}{\partial x_j} + \frac{\partial \bar{u}_k}{\partial x_j}\frac{\partial u_j'}{\partial x_k} - \frac{2}{3}\frac{\partial \bar{u}_k}{\partial x_k}\frac{\partial u_j'}{\partial x_j}\right]\right. \\ &\quad \left. + \mu'\left[\left(\frac{\partial \bar{u}_k}{\partial x_j}\right)^2 + \frac{\partial \bar{u}_k}{\partial x_j}\frac{\partial \bar{u}_j}{\partial x_k} - \frac{2}{3}\frac{\partial \bar{u}_k}{\partial x_k}\frac{\partial \bar{u}_j}{\partial x_j}\right]\right\}\end{aligned} \tag{D3}$$

$$\rho' = \frac{p'\gamma}{\bar{T}} - \frac{\gamma \bar{p}.T'}{\bar{T}^2} \tag{D4}$$

Now, for clarity, the notation u, v, x, y is used. Assuming a perturbation in the form $\phi' = \hat{\phi}(y)\exp(\mathrm{i}(\alpha x - \Omega t))$ and exploiting the spatial homogeneity of the base flow, the perturbed governing equations are given in terms of $\hat{u}, \hat{v}, \hat{p}, \hat{T}$ by

$$\mathrm{i}\Omega\hat{p} - \mathrm{i}\Omega\bar{p}\hat{T}/\bar{T} = \mathrm{i}\alpha\bar{p}\hat{u} + \left(\bar{p}\frac{d}{\mathrm{d}y} + \frac{\bar{T}}{\gamma}\frac{\mathrm{d}\bar{\rho}}{\mathrm{d}y}\right)\hat{v} + \mathrm{i}\alpha\bar{u}\hat{p} - \frac{\bar{\rho}\bar{u}}{\gamma}\hat{T} \tag{D5}$$

$$\begin{aligned}-\mathrm{i}\Omega\bar{\rho}\hat{u} - \frac{1}{Re_a}\left[\bar{\mu}\frac{\mathrm{d}^2}{\mathrm{d}y^2} + \frac{\mathrm{d}\bar{\mu}}{\mathrm{d}\bar{T}}\frac{\mathrm{d}\bar{T}}{\mathrm{d}y}\frac{\mathrm{d}}{\mathrm{d}y}\right]\hat{u} + \bar{\rho}\frac{\mathrm{d}\bar{u}}{\mathrm{d}y}\hat{v} \\ - \frac{1}{Re_a}\left[\frac{\mathrm{d}\bar{\mu}}{\mathrm{d}\bar{T}}\frac{\mathrm{d}^2\bar{u}}{\mathrm{d}\bar{T}^2} + \frac{\mathrm{d}\bar{\mu}}{\mathrm{d}\bar{T}}\frac{\mathrm{d}\bar{u}}{\mathrm{d}y}\frac{\mathrm{d}}{\mathrm{d}y} + \frac{\mathrm{d}^2\bar{\mu}}{\mathrm{d}\bar{T}^2}\frac{\mathrm{d}\bar{T}}{\mathrm{d}y}\frac{\mathrm{d}\bar{u}}{\mathrm{d}y}\right]\hat{T} \\ = -\left[\mathrm{i}\alpha\bar{\rho}\bar{u} + \frac{4}{3}\frac{1}{Re_a}\bar{\mu}\alpha^2\right]\hat{u} + \frac{\mathrm{i}\alpha}{Re_a}\left[\frac{1}{3}\bar{\mu}\frac{\mathrm{d}}{\mathrm{d}y} + \frac{\mathrm{d}\bar{\mu}}{\mathrm{d}\bar{T}}\frac{\mathrm{d}\bar{T}}{\mathrm{d}y}\right]\hat{v} - \mathrm{i}\alpha\hat{p}\end{aligned} \tag{D6}$$

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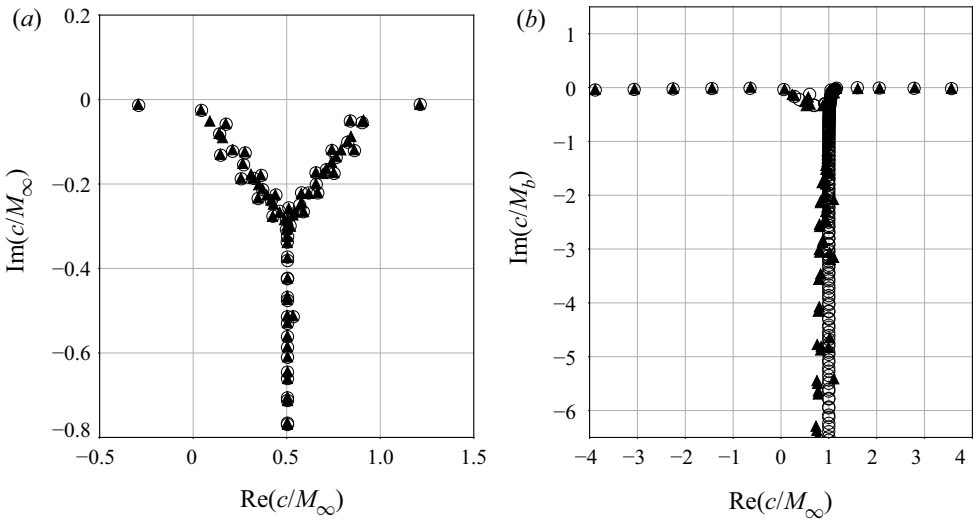

Figure 23. Complex wave speed of compressible Couette flow at $M_\infty = 2.0, Re_\infty = 2 \times 10^5$ (*a*) and supersonic channel flow at $M_b = 3.0, Re_b = 4880$ (*b*). Symbols are: (▲) reference data from Hu & Zhong (1997) for (*a*) and Friedrich & Bertolotti (1997) for (*b*); (unfilled circles) data from current solver.

$$-\mathrm{i}\Omega\bar{\rho}\hat{v} - \frac{1}{Re_a}\frac{4}{3}\left[\bar{\mu}\frac{\mathrm{d}^2}{\mathrm{d}y^2} + \frac{\mathrm{d}\bar{\mu}}{\mathrm{d}\bar{T}}\frac{\mathrm{d}\bar{T}}{\mathrm{d}y}\frac{\mathrm{d}}{\mathrm{d}y}\right]\hat{v} + \frac{\mathrm{d}\hat{p}}{\mathrm{d}y}$$
$$= \frac{\mathrm{i}\alpha}{Re_a}\left[\frac{1}{3}\bar{\mu}\frac{\mathrm{d}}{\mathrm{d}y} - \frac{2}{3}\frac{\mathrm{d}\bar{\mu}}{\mathrm{d}\bar{T}}\frac{\mathrm{d}\bar{T}}{\mathrm{d}y}\right]\hat{u} - \left[\mathrm{i}\alpha\bar{\rho}\bar{u} + \frac{1}{Re_a}\bar{\mu}\alpha^2\right]\hat{v} + \frac{\mathrm{i}\alpha}{Re_a}\frac{\mathrm{d}\bar{\mu}}{\mathrm{d}\bar{T}}\frac{\mathrm{d}\bar{T}}{\mathrm{d}y}\hat{T} \tag{D7}$$

$$(\gamma-1)\mathrm{i}\Omega\hat{p} - \mathrm{i}\Omega\bar{p}\hat{T} - \frac{2(\gamma-)\bar{\mu}}{Re_a}\frac{\mathrm{d}\bar{u}}{\mathrm{d}y}\frac{\mathrm{d}\hat{u}}{\mathrm{d}y} + \bar{\rho}\frac{\mathrm{d}\bar{T}}{\mathrm{d}y}\hat{v}$$
$$-\frac{1}{PrRe_a}\left[\bar{k}\frac{\mathrm{d}^2}{\mathrm{d}y^2} + 2\frac{\mathrm{d}\bar{k}}{\mathrm{d}\bar{T}}\frac{\mathrm{d}\bar{T}}{\mathrm{d}y}\frac{\mathrm{d}}{\mathrm{d}y} + \frac{\mathrm{d}^2\bar{k}}{\mathrm{d}\bar{T}^2}\left(\frac{\mathrm{d}\bar{T}}{\mathrm{d}y}\right)^2 + \frac{\mathrm{d}\bar{k}}{\mathrm{d}\bar{T}}\frac{\mathrm{d}^2\bar{T}}{\mathrm{d}y^2}\right]\hat{T} - \frac{(\gamma-1)}{Re_a}\frac{\mathrm{d}\bar{\mu}}{\mathrm{d}\bar{T}}\left(\frac{\mathrm{d}\bar{u}}{\mathrm{d}y}\right)^2\hat{T}$$
$$= \mathrm{i}\alpha\frac{2(\gamma-1)}{Re_a}\bar{\mu}\frac{\mathrm{d}\bar{u}}{\mathrm{d}y}\hat{v} + \mathrm{i}\alpha(\gamma-1)\bar{u}\hat{p} - \left[\mathrm{i}\alpha\bar{\rho}\bar{u} + \frac{1}{PrRe_a}\bar{k}\alpha^2\right]\hat{T}. \tag{D8}$$

Note that the perturbation of dynamic viscosity and thermal conductivity is related to the perturbed temperature via

$$\mu' = \frac{\mathrm{d}\bar{\mu}}{\mathrm{d}\bar{T}}T', \quad k' = \frac{\mathrm{d}\bar{k}}{\mathrm{d}\bar{T}}T'. \tag{D9a,b}$$

The above equations are solved at Gauss–Lobatto–Chebyshev points. To validate the solver, a supersonic Couette flow case from Hu & Zhong (1997) and the supersonic channel flow case from Friedrich & Bertolotti (1997) are tested and compared to the literature, as shown in figure 23. Good agreement has been achieved for both cases.

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